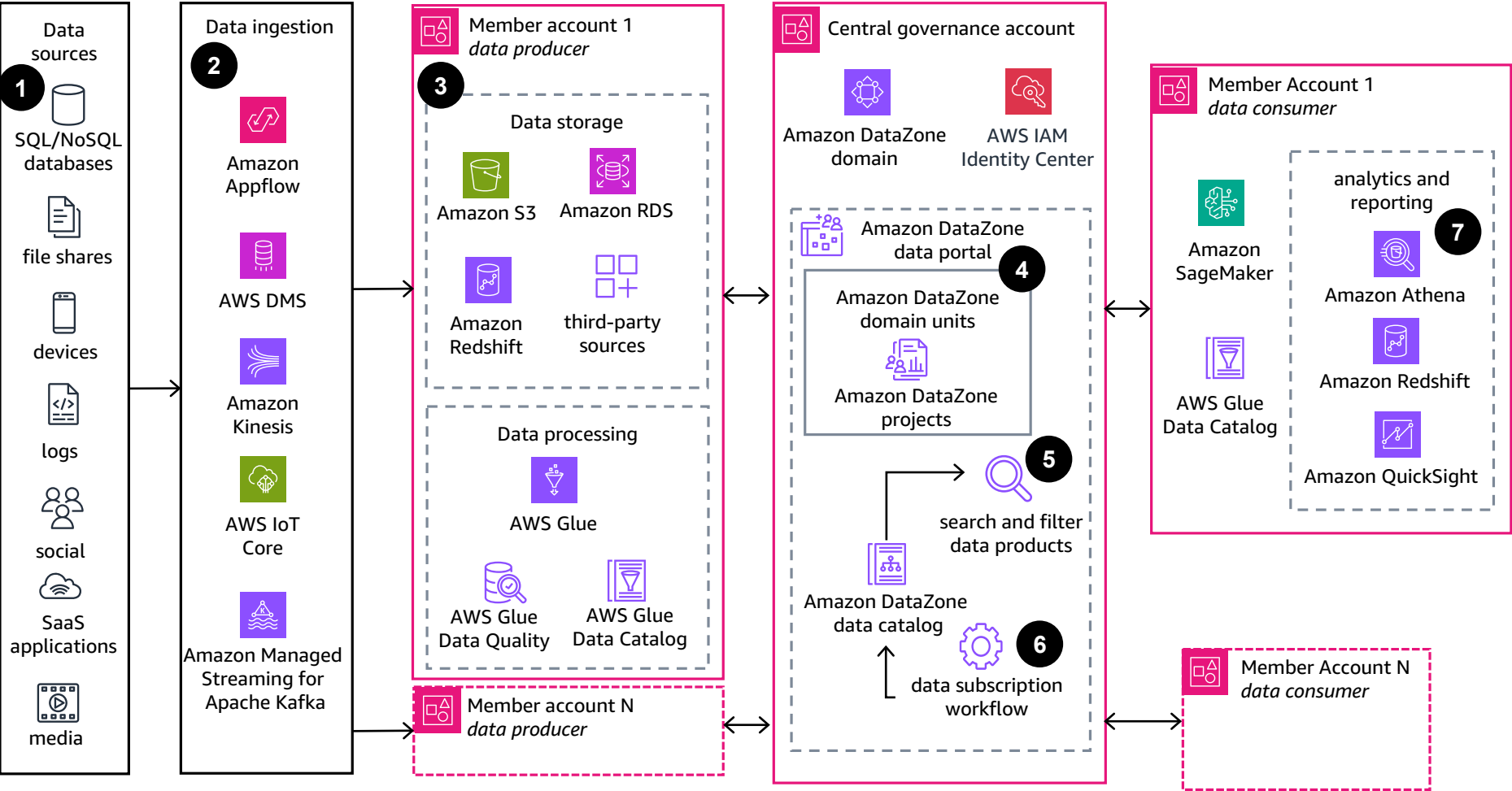


Data Mesh Reference Architecture with Amazon DataZone

Data Mesh is a decentralized architectural and organizational framework that helps organizations accelerate innovation and drive business value. This reference architecture shows how Amazon Web Services customers can leverage Amazon DataZone to build a data mesh-based data solution.



- 1 Data is gathered from data sources across the enterprise through databases, file shares, edge devices, logs, social networks, SaaS applications, and streaming media.
- 2 Based on the source system and end user requirements, raw data is ingested into **Amazon AppFlow**, **AWS Database Migration Service**, **Amazon Kinesis**, **AWS IoT Core**, and **Amazon Managed Streaming for Apache Kafka**.
- 3 In the producer account, raw data is transformed using **AWS Glue**. Metadata is stored in **AWS Glue Catalog**, data quality is measured using **AWS Glue Data Quality**. Data stored in **Amazon Simple Storage Service (Amazon S3)**, **Amazon Redshift**, **Amazon Relational Database Service**, and third-party sources is registered as an asset in the **Amazon DataZone** catalog hosted in the central governance account using **Amazon DataZone** data source jobs.
- 4 The central governance account hosts the **Amazon DataZone** domain and the related data portal. The AWS accounts of the data producers and consumers are associated with the **Amazon DataZone** domain. Amazon DataZone projects belonging to the data producers and consumers are created under related Amazon DataZone domain units.
- 5 End users of assets log into the **Amazon DataZone** data portal hosted in the central governance account using their **AWS Identity and Access Management (IAM)** credentials or single sign-on (SSO) integration through **IAM Identity Center**. They search, filter, and view asset information like data quality, business, and technical metadata.
- 6 When the end users find assets of interest, they request access using the subscription feature of **Amazon DataZone**. Based on the validity of the request, the asset owner approves or rejects the request.
- 7 After the subscription request is granted and fulfilled, the asset is accessed in the consumer account for AI/ML model development using **Amazon SageMaker**; for analytics and reporting, use **Amazon Athena**, **Amazon Redshift**, and **Amazon QuickSight**.

